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WHO INFLUENZA CENTRE

LONDON

Report
February 2009

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Acknowledgements

We thank all who have contributed information and viruses, and associated data, to the WHO Global Influenza Surveillance Network, which provide the basis for our current understanding of recently circulating influenza viruses, and this brief summary. In addition to data shared among the other 3 WHO Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Austria, Denmark, Finland, France, Germany, Hong Kong, Italy, Netherlands, Norway, Portugal, South Africa, Sweden, Switzerland and UK.

Influenza activity, October 2008 to March 2009

Influenza was in general mild. Influenza activity in most parts of the world was lower compared with the same period last year, while in Europe it was higher than during the previous two seasons (Figure 1).
In the southern hemisphere, influenza activity continued into November; influenza B viruses had predominated in Australia and New Zealand during September, while influenza A(H1N1), A(H3N2) and B viruses circulated in different countries to varying extents.
In Europe, influenza activity commenced in October (Figure 1) and followed a west to east progression. A(H3N2) viruses predominated for most of the season, with influenza B viruses increasing during February and March. H3N2 viruses also predominated in Japan, whereas A(H1N1) viruses were relatively more prevalent in other Asian countries, such as Korea. In north African countries A(H3N2) and A(H1N1) viruses co-circulated, while in some west African countries A(H3N2) or B viruses predominated. In North America A(H1N1)viruses predominated in the USA, while B viruses predominated in Canada.

Antigenic and genetic characteristics of influenza A(H1N1), A(H3N2) and B viruses isolated during October 2008 to February 2009.

Of approximately 550 human influenza viruses, isolated in 33 countries during October 2008 to February 2009, the majority(68%) were A(H3N2) and 20% were A(H1N1) (Table 1). Influenza B viruses accounted for 12% of the viruses received, including similar numbers of B/Victoria lineage and B/Yamagata lineage viruses; viruses from African countries were of the B/Yamagata/16/88 lineage, whereas most of those from European countries were of the B/Victoria/2/87 lineage.

A(H1N1) viruses

In **HI tests**, the majority of H1N1 viruses were antigenically closely related to the MDCK cell-grown reference viruses A/Netherlands/345/2007 (clades 2B), A/Seychelles/2239/2008 (clade 2B), A/Hong Kong/1856/2008 (clade 2C) and/or A/Hong Kong/1870/2008 (clade 2C) (Table 2). Antisera to these viruses did not distinguish between clade 2B and 2C HAs. HI titres with antisera to the egg-grown vaccine virus A/Brisbane/59/07 were in general reduced relative to the homologous titre, apparently reflecting differences between the HAs (e.g. an amino acid substitution at position 186, D186N/V) of cell culture and egg isolates. No A/New Caledonia/20/99-like viruses were identified.

The majority of **HA and NA sequences** fell within clade 2B, represented by A/Brisbane/59/2007 (Figures 2 and 3). Most clade 2B HA sequences of recent isolates possess the common amino acid change A189T, but fall into subgroups distinguished by different signature amino acid changes S141N+G185A, S141R+N183D, N183S+G185S/N (+N244S), G185V+H192R or H192N (Figure 2, Table 3). NA sequences fell into corresponding subgroups within clade 2B (based on nucleotide sequences, but were not distinguished by amino acid differences) (Figure 3). The majority of NAs possessed the H275Y mutation, conferring oseltamivir resistance, and the D354G substitution (Figure 3).

Relatively few HA and NA sequences (mainly of viruses from China) fell within clade 2C, represented by A/Hong Kong/1856/2008.

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Antiviral susceptibility

The majority of H1N1 viruses tested were shown to be resistant to oseltamivir, either by enzyme inhibition assay or the presence of the H275Y mutation in NA, or both (Table1). Data produced by the EISS/VIRGIL network showed that 97% of European isolates tested were resistant to oseltamivir, but retained sensitivity to zanamivir and amantadine/rimantadine.

A(H3N2) viruses

There was a marked increase in the proportion of H3N2 viruses which failed to agglutinate turkey or human red blood cells (RBCs), but agglutinated guinea pig RBCs, such that the majority of recent isolates received from some countries agglutinated only guinea pig RBCs.

As previously, various **HI tests** using **turkey RBCs** (Table 4) continued to show that most H3N2 viruses gave patterns of reduced HI titres similar to those observed for MDCK cell-grown reference viruses, such as A/Trieste/25c/2007 and/or A/Johannesburg/15/2008, which also gave relatively low homologous HI titres with their corresponding ferret antisera, compared with egg-passaged reference viruses. Furthermore, antisera raised in ferrets infected with MDCK cell-grown or egg-grown reference viruses gave similar patterns of differential HI reactivity with the various reference viruses, indicating that the cell culture-grown reference viruses and egg-grown vaccine viruses elicited similar antibody responses in ferrets.

HI tests of viruses which failed to agglutinate turkey (or human) RBCs, with **guinea pig RBCs** (Table 5) gave high HI titres, comparable to those obtained with egg-grown reference viruses and showed that most of the viruses were antigenically closely related to the vaccine strains A/Brisbane/10/2007 and A/Uruguay/716/2007.

Neutralisation tests did not distinguish between egg-grown (e.g. A/Brisbane/10/2007) and MDCK cell-grown (e.g. A/Johannesburg/15/2008) reference viruses. They showed that for several of the viruses which gave relatively low HI titres (blue in Table 4), as for some which agglutinated only guinea pig RBCs (blue in Tables 5 and 6), neutralization titres of reference antisera were similar to those with the homlogous reference viruses, indicating that the viruses were antigenically closely related to the A/Brisbane/10/2007 and A/Uruguay/716/2007 vaccine strains (Table 6).

The majority of **HA sequences** fell within the phylogenetic group characterised by the change K173Q relative to the sequence of A/Brisbane/10/2007 (Figure 4). Sequences were fairly homogeneous and no prominent phylogenetic subgroups were distinguished by additional common amino acid substitutions.

The majority of **NA sequences** fell within a corresponding group defined by the substitutions D147N and I215V, relative to the sequence of A/Brisbane/10/2007 (Figure 5). The sequences of many recent European isolates fell within a subgroup characterized by the additional changes P386H and I464L (the corresponding HA sequences did not possess any common amino acid differences).

Antiviral susceptibility

All viruses analysed carried asparagine at position 31 in M2, indicative of resistance to amantadine and rimantadine, and were sensitive to oseltamivir and zanamivir.

B viruses

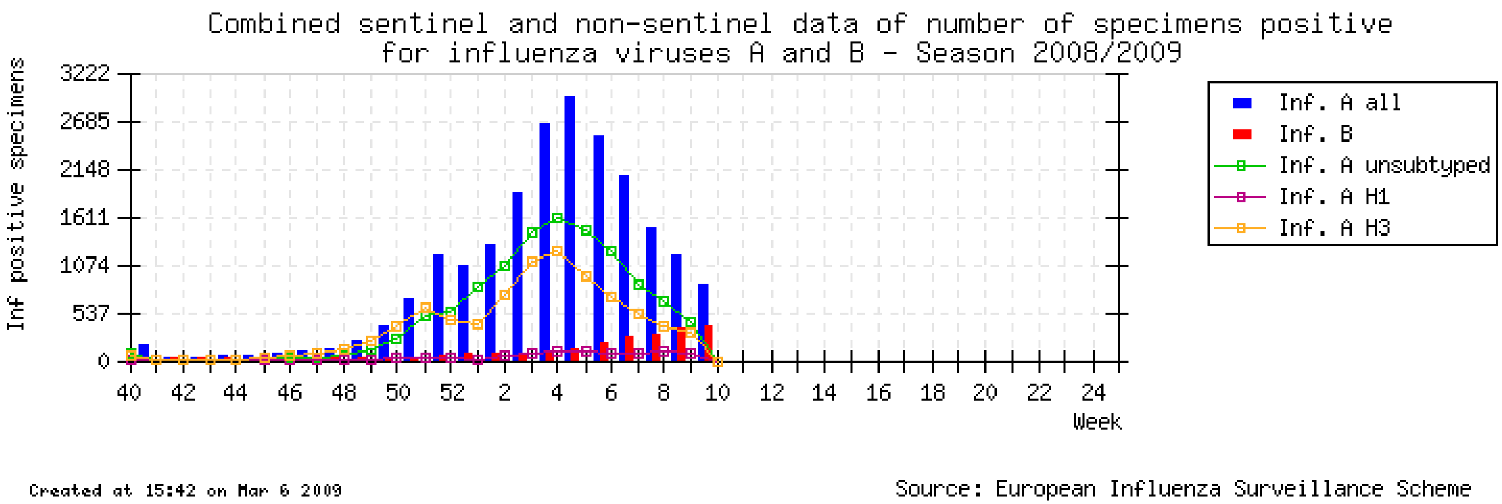
In **HI tests**, the majority of **B/Victoria-lineage viruses** were shown to be antigenically closely related either to the MDCK cell-grown reference virus B/Hong Kong/45/2005 (B/Malaysia/2506/2004-like) or to B/Brisbane/60/2008 and B/Brisbane/33/2008, the strains recommended by WHO for inclusion in the 2009-2010 vaccine (Table 7). Reduced HI titres, relative to those with the homologous egg-grown reference viruses reflect differences commonly observed between the reactivities of egg- and MDCK cell-grown viruses.

The majority of **HA sequences** of recent isolates fell within clade 2, characterized by the changes N75K, V146I, N165K and N172P relative to the HA of B/Malaysia/2506/2004 and represented by B/Brisbane/60/2008 and B/Brisbane/33/2008 (Figure 6, Table 9).
NA sequences of these viruses fell within clade 2A and were characterized by additional changes I204V, D329N and A358E, relative to the sequence of B/Victoria/304/2006 (Figure 7, Table 9). NA sequences of some recent isolates with clade 1 HAs, such as B/Brazil/2937/2008, fell within clade 2B (Figure 7).

HI tests of the **B/Yamagata-lineage viruses** showed that many were antigenically closely related to the vaccine strains, B/Florida/4/2006 and B/Brisbane/3/2007, whereas others were more closely related to the recent reference viruses, B/Barcelona/143/2008 (MDCK), B/England/145/2008 (egg), B/Valladolid/18/2008 (MDCK) and/or B/Bangladesh/3333/2007 (egg), (Table 8).

HA sequences of recent B/Yamagata-lineage isolates were similar to those of viruses isolated during 2007 and 2008 and fell into one of the 3 principal clades, with most falling within clade 3, represented by e.g. B/Bangladesh/3333/2007 and B/England/145/2008 (Figure 8, Table 9). HAs in the three clades were not distinguished in HI tests. **NA sequences** fell into corresponding phylogenetic subgroups, as previously; those represented by the vaccine viruses (clades 1 and 2) were similar and distinguished from those of clade 3 by 7 amino acid differences, including the substitutions D463N and A465T, relative to the sequence of B/Bangladesh/3333/2007 (Figure 7, Table 9).

Figure 1: Influenza situation in Europe 2008-09 season (weeks 40/08-09/09)



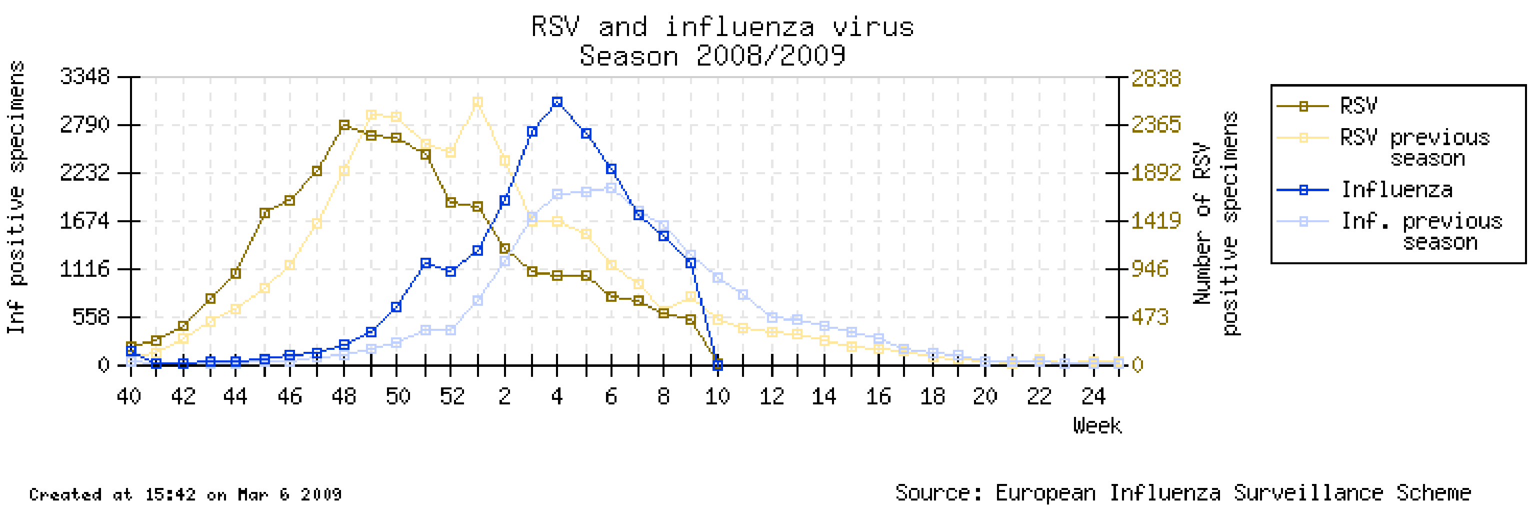


Table 1. Human influenza isolates, October 2008 - February 2009

Country	H1N1	H3N2	B Yamagata lineage	B Victoria lineage
October				
Argentina	1	1		
Cameroon	1 1		3	
France		5		
Germany		1	1	2
Ghana	1 1		4	
Hong Kong	1	2	1	
Madagascar	2	2		
Norway		1		
Spain			2	1
Sweden		4		
UK	3 3			1
November				
Algeria		1		
Cameroon	9 9		6	
Belgium		1		1
Denmark		2		
Egypt		6		
France	2 2	10	1	
Germany		2	1	
Ghana			7	
Hong Kong	2 2	1		
Italy		6	1	
Madagascar		2		
Mauritius		1		
Norway	1 1	4	1	
Portugal		1		
Romania		1		
Singapore	1 1			1
Spain				4
Sweden	1 1	2		
Switzerland		1		
UK	1 1			1
December				
Algeria		8		
Belgium		22		1
Cameroon	4 4		1	
Finland		6		
France	9 9	10		
Germany	4 4	13		5
Ghana			3	
Israel	5 5	3		
Italy		16		
Korea	9 9	1		
Latvia		4		
Luxembourg	1 ?	4		
Morocco	6 6			3
Norway	4 4	11		
Portugal	4 ?	12		
Singapore	1 1			
Slovakia		1		
Spain		34		
Sweden	2 2	5		
Switzerland		10	1	
UK	1 1	12		2
January 2008				
Belgium		50	1	1
Czech Republic	3 ?	12		
Egypt			1	
Germany		2		
Greece		8		3
Iran	2 2			
Israel	4 ?	8		3
Italy	5 ?	14		
Latvia		2		
Morocco	15 15	7		
Norway		2		
Portugal	2 ?	1		
Romania		9		
Slovakia	1 ?	6		1
Slovenia		3		
Spain		11		
Switzerland		9	1	1
Tunisia		2		
February 2009				
Czech Republic	1 ?			2
Total = 553	**109**	**375**	**36**	**33**
	20%	**68%**	**69 (12%)**	

oseltamivir resistant; ? = yet to be determined

Table 2. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret sera A/NC 20/99 F19/08	A/HK 2652/06 F14/06	A/SI 3/06 F8/07	A/Neth 345/07 F23/07	A/Bris 59/07 F6/08	A/SP 12/08 F17/08	A/HK 1856/08 F28/08	A/Sey 2239/08 F27/08	A/HK 1870/08 F4/09
REFERENCE VIRUSES											
A/New Caledonia/20/99	06/09/1999	Ex	**640**	<	40	80	40	40	160	80	<
A/Hong Kong/2652/2006	17/07/2006	E2 \2	40	**320**	320	160	320	320	640	160	1280
A/Solomon Islands/3/2006	21/08/2006	E3 \2	160	320	**640**	640	160	160	1280	640	320
A/Netherlands/345/2007	12/10/2007	MDCK2 \4	80	160	320	**1280**	160	320	2560	640	160
A/Brisbane/59/2007	01/07/2007	E2 \1	40	80	320	160	**320**	320	320	160	640
A/St.Petersburg/12/2008	13/02/2008	E2 \2	40	80	320	160	320	**640**	320	160	640
A/Hong Kong/1856/2008	24/07/2008	MDCK2 \2	40	80	40	160	<	40	**1280**	160	320
A/Seychelles/2239/2008	03/06/2009	MDCK1 \2	40	80	80	320	80	160	1280	**320**	320
A/Hong Kong/1870/2008	26/07/2008	MDCK4 \2	<	160	160	80	160	160	320	80	**640**
TEST VIRUSES											
A/England/556/2008	29/11/2008	MDCK P1 \1	40	40	80	160	<	80	320	160	ND
A/Singapore/43/2008	03/12/2008	MDCK1 \1	<	40	80	160	160	160	160	320	160
A/Lisboa/29/2008	03/12/2008	MDCK \2	40	160	320	640	320	160	1280	640	320
A/Incheon/4532/2008	16/12/2008	MDCK-P2 \1	<	80	160	80	160	160	80	80	320
A/Luxembourg/1344/2008	unknown	MDCKx \1	40	160	80	320	80	160	1280	320	ND
A/Gyeonggibuk/4500/2008	15/12/2008	MDCK-P2 \1	<	80	80	80	160	160	80	80	320
A/Incheon/4532/2008	16/12/2008	MDCK-P2 \1	<	80	160	80	160	160	80	80	320
A/Hong Kong/3176/2008	15/11/2008	MDCK2 \1	<	40	40	320	80	160	160	320	160
A/Paris/781/2008	15/12/2008	C1 \1	40	160	160	1280	80	160	640	1280	320
A/Rabat/109/09	01/12/2008	Cx \3	<	<	40	160	<	80	640	160	160
A/Marrakech/510/2009	12/01/2009	Cx \2	<	80	40	320	40	80	640	160	160
A/Berlin/59/2008	18/12/2008	MDCK4 \1	<	80	80	160	40	80	320	160	80
A/Sachsen-Anhalt/54/2008	22/12/2008	MDCK4 \1	<	80	80	80	40	80	160	80	160
A/Parma/25/09	12/01/2009	MDCK1 \1	<	80	160	80	80	80	160	80	160
A/Parma/27/09	14/01/2009	MDCK1 \1	<	40	<	160	40	80	320	160	80
A/Fes/213/09	14/01/2009	Cx \1	40	80	80	320	40	160	1280	320	160
A/Israel/63/08	15/12/2008	MDCKx \1	<	40	80	80	80	160	160	160	160
A/Israel/64/08	19/12/2008	MDCKx \1	40	80	160	320	80	160	640	640	320
A/Norway/2262/2008	08/12/2008	MDCK1 \1	40	160	80	320	80	160	320	640	160
A/Sweden/8/08	09/12/2008	MDCK1 \1	40	80	160	640	40	160	1280	640	ND
A/Pilsen/44/2009	15/01/2009	MDCK3 \1	<	40	40	80	80	160	80	40	160
A/Prague/89/2009	02/02/2009	MDCK2 \1	<	80	40	80	80	160	320	80	160
A/Cameroon/08-224/2008	04/12/2008	MDCKx \1	<	80	80	160	80	160	160	160	160
A/Cameroon/08-232/2008	05/12/2008	MDCKx \1	40	80	160	40	80	320	80	80	320
A/Laayoune/208/09	17/01/2009	Cx \1	<	80	80	40	160	320	80	40	320

1. < = <40; ND = not done

Vaccine strain

Figure 2. Phylogenetic comparison of influenza H1N1 HA genes

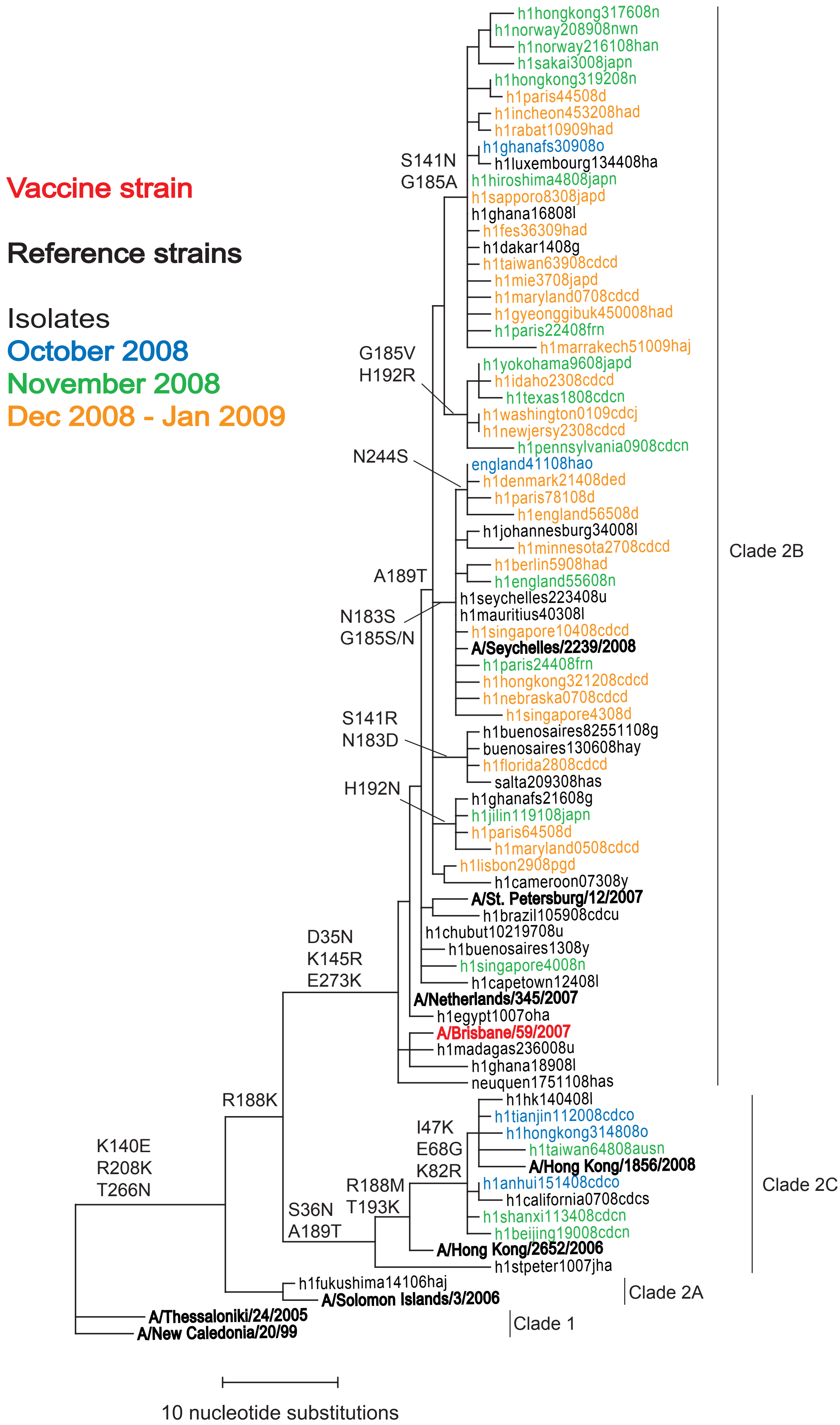


Figure 3. Phylogenetic comparison of H1N1 neuraminidase genes

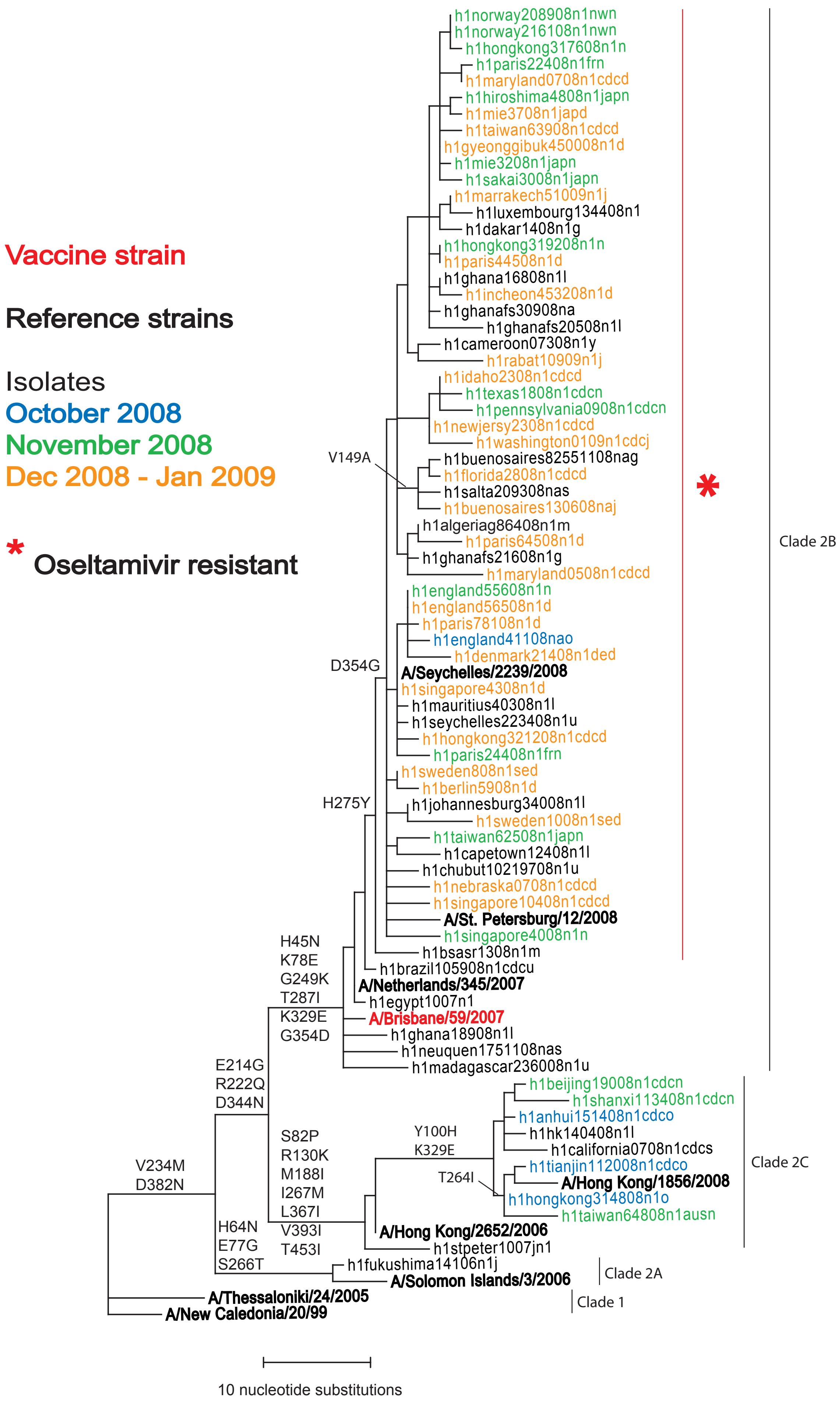


Table 3. Amino acid changes characteristic of the recent principal H1N1 sequence variants

Clade	Representative strain	Amino acid changes[1]	
		HA	NA[2]
2B	A/Seychelles/2239/2008 (A/Brisbane/59/2007)	D35N K145R R188K E273K A189T (S141N + G185A)* (N183S + G185S/N)* (S141R + N183D)* (G185V + H192R)* (H192N)* (N244S)*	H45N K78E G249K T287I K329E H275Y (V149A)
2C	A/Hong Kong/1856/2008	S36N I47K E68G K82R R188M A189T T193K	S82P Y100H R130K M188I I267M K329E L367I V393I T453I (T264I)

1. Relative to A/Solomon Islands/3/2006 (Clade 2A)
2. E214G, R222Q and D344N are common to both 2B and 2C
* representative of recent subgroups

Table 4. Antigenic analyses of influenza A H3N2 viruses (Turkey RBCs)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret sera A/Wis 67/05 F18/08	A/Trieste 25c/07 F5/07	A/Wis 3/07 F15/07	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/JHB 15/08 F22/08	A/Eng 394/08 F32/08	A/Bris 24/08 F3/09
REFERENCE VIRUSES										
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 \3	**5120**	5120	1280	5120	5120	2560	320	5120
A/Trieste/25c/2007	Jan-07	MDCKx \2	<	**80**	160	320	640	80	160	160
A/Wisconsin/3/2007	21/01/2007	Ex \2	5120	5120	**5120**	5120	5120	2560	1280	5120
A/Brisbane/10/2007	06/02/2007	E2 \3	2560	5120	640	**5120**	5120	2560	320	5120
A/Uruguay/716/2007	21/06/2007	spfck1, E3 \1	2560	2560	640	2560	**5120**	1280	320	2560
A/Johannesburg/15/2008	25/06/2008	MDCKx \2	<	160	160	320	320	**160**	160	160
A/England/394/2008	22/09/2008	MDCK2 \7	80	320	320	640	320	320	**640**	320
A/Brisbane/24/2008	23/06/2008	E5 \1	2560	5120	2560	5120	5120	5120	640	**5120**
TEST VIRUSES										
A/Madagascar/3226/2008	23/09/2008	MDCK1 \1	<	40	80	40	80	40	80	ND
A/Netherlands/379/2008	24/09/2008	MK2 \1	<	<	80	40	40	<	40	ND
A/Norway/2118/2008	25/11/2008	MDCK1 \1	<	80	80	80	160	ND	80	ND
A/Thuringen/76/2008	26/11/2008	1.Pa(RKI) \1	<	80	80	40	80	ND	80	ND
A/Luxembourg/1325/2008	02/12/2008	MDCKx \1	80	320	320	160	1280	160	160	ND
A/England/561/2008	04/12/2008	MDCK P1 \1	<	40	80	40	40	40	40	ND
A/England/572/2008	08/12/2008	MDCK P1 \1	<	40	80	40	40	<	40	ND
A/Finland/286/2008	08/12/2008	MDCK2 \1	<	80	160	80	320	40	80	ND
A/England/85060527/2008	10/12/2008	SIAT P2 \1	<	40	160	40	40	40	80	ND
A/Paris/130/2008	20/10/2008	C2 \1	40	160	160	320	320	320	160	160
A/Denmark/191/2008	13/11/2008	MDCK3 \1	40	80	160	160	160	160	80	160
A/Toulouse/1317/08	14/11/2008	P3MDCK \1	<	80	160	160	160	80	80	80
A/Egypt/59/2008	16/11/2008	C1 \1	<	80	80	80	160	80	40	80
A/Hong Kong/3197/2008	26/11/2008	MDCK2 \1	80	160	160	640	320	160	320	160
A/Sweden/6/08	08/12/2008	MDCK2 \1	80	160	160	320	320	160	160	160
A/Norway/2308/2008	15/12/2008	MDCK1 \1	80	160	320	320	320	160	160	160
A/England/669/2008	30/12/2008	MDCK P1 \1	640	640	320	1280	2560	640	640	320
A/Norway/1/2009	30/12/2008	MDCK1 \1	1280	2560	1280	2560	5120	1280	1280	640
A/Kanagawa/65/2008	unknown	MDCK2+2 \1	<	80	160	160	160	80	160	80
A/Taiwan/471/2008	unknown	E3+2 \1	320	160	320	640	640	320	160	1280
A/Stockholm/26/2008	12/11/2008	MDCK1 \1	640	640	640	1280	1280	640	320	640
A/Norway/2264/2008	09/12/2008	MDCK1 \1	<	40	80	160	80	40	80	40
A/Baden-Württemburg/199/2008	15/12/2008	MDCK4 \2	<	80	80	160	160	80	80	80
A/Bayern/47/2008	15/12/2008	MDCK5 \1	<	80	80	160	160	80	80	80
A/Niedersachsen/107/2008	15/12/2008	MDCK2 \1	<	80	160	320	160	80	80	160
A/Sweden/9/2008	17/12/2008	MDCK2 \1	640	1280	640	640	2560	640	160	640
A/Baden-Württemburg/196/2008	18/12/2008	MDCK3 \1	<	160	160	320	320	160	160	160
A/Niedersachsen/110/2008	18/12/2008	MDCK4 \1	80	320	640	640	640	640	160	640
A/Berlin/62/2008	22/12/2008	MDCK4 \1	<	80	80	160	80	80	80	80
A/Firenze/24/08	22/12/2008	I MDCK \1	<	80	160	160	160	80	160	160
A/Firenze/26/08	22/12/2008	I MDCK \1	<	80	160	320	160	160	160	160
A/Switzerland/1397477/2008	26/12/2008	MDCK1 \1	<	80	160	160	160	80	160	80
A/Switzerland/1397481/2008	26/12/2008	MDCK2 \1	<	40	80	160	40	40	80	40
A/Parma/20/09	30/12/2008	MDCK1 \1	<	80	160	320	160	160	160	80
A/Latvia/1-34/2009	02/01/2009	MDCK1 \2	<	80	160	160	80	80	80	80
A/Bremen/1/2009	05/01/2008	MDCK3 \1	80	160	160	320	320	160	160	80
A/Slovenia/51/2009	05/01/2009	MDCKx \1	<	80	160	320	160	80	160	160
A/Slovenia/55/2009	07/01/2009	MDCKx \1	<	80	80	160	80	80	80	80
A/Parma/22/09	10/01/2009	MDCK1 \1	<	40	160	160	80	80	160	80
A/Parma/17/09	12/01/2009	MDCK1 \1	<	80	160	160	160	160	160	160
A/Lisboa/04/2009	30/12/2008	MDCK2 \2	<	40	40	80	80	40	40	40
A/Cluj/34/2009	06/01/2009	MDCK3 \2	<	40	80	160	80	40	80	80
A/Barcelona/321/2009	09/12/2008	MDCK2 \2	<	40	160	160	80	80	80	80
A/Barcelona/328/2009	16/12/2008	MDCK2 \2	<	40	160	160	80	80	80	80
A/Algeria/G208/2009	27/12/2008	Cx \1	<	40	80	80	160	80	80	80
A/Bucuresti/1816/2008	01/11/2008	MDCK 3 \1	<	40	80	80	80	40	80	80
A/Lasi/22/2009	07/01/2009	MDCK 3 \1	<	80	160	160	80	80	160	160
A/Cluj/35/2009	09/01/2009	MDCK 3 \1	<	80	160	160	80	80	160	160
A/Bucuresti/54/2009	12/01/2009	MDCK 2 \1	<	40	80	160	80	80	80	80
A/Israel/9/09	10/01/2009	MDCKx \1	<	320	320	640	320	320	320	640
A/Israel/13/09	17/01/2009	MDCKx \1	<	<	80	160	80	40	80	40
A/Barcelona/329/2009	16/12/2008	MDCK2 \1	<	<	80	160	40	80	160	80
A/Barcelona/V359A/2009	13/01/2009	MDCK1 \1	<	40	80	80	80	80	80	80
A/Algeria/G216/2009	30/12/2008	Cx \1	<	80	160	160	80	80	160	160
A/Algeria/G217/2009	30/12/2008	Cx \1	<	40	80	80	80	40	80	80
A/Firenze/14/08	27/11/2008	I MDCK \2	<	80	80	160	160	80	80	80
A/Bratislava/148/2009	21/01/2009	1 \1	80	160	160	320	320	80	160	160
A/Bratislava/151/2009	20/01/2009	1 \1	320	160	320	320	320	160	160	160
A/Belgium/G318/2009	12/01/2009	MDCK1 \3	<	40	80	160	80	40	40	80
A/Belgium/G344/2009	13/01/2009	MDCK1 \3	<	80	80	160	80	80	80	160

1. < = <40; ND = not done

Vaccine strains

Also tested by neutralisation

Table 5. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBCs)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret sera A/Wis 67/05 F18/08	A/Trieste 25c/07 F5/07	A/Wis 3/07 F15/07	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/JHB 15/08 F22/08	A/Eng 394/08 F32/08	A/Bris 24/08 F3/09
REFERENCE VIRUSES										
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 \3	**1280**	1280	640	1280	2560	640	80	1280
A/Trieste/25c/2007	Jan-07	MDCKx \2	80	**160**	160	320	160	160	160	160
A/Wisconsin/3/2007	21/01/2007	Ex \2	5120	5120	**2560**	5120	5120	2560	640	5120
A/Brisbane/10/2007	06/02/2007	E2 \3	1280	2560	640	**1280**	2560	1280	320	2560
A/Uruguay/716/2007	21/06/2007	spfck1, E3 \1	1280	2560	640	1280	**2560**	1280	320	1280
A/Johannesburg/15/2008	25/06/2008	MDCKx \2	160	320	320	320	320	**320**	320	640
A/England/394/2008	22/09/2008	MDCK2 \7	160	320	320	320	640	320	**640**	320
A/Brisbane/24/2008	23/06/2008	E5 \1	2560	2560	2560	5120	5120	2560	320	**5120**
TEST VIRUSES										
A/England/392/2008	11/09/2008	C1 \5	80	320	160	640	640	80	1280	ND
A/England/394/2008	22/09/2008	C2 \5	160	320	160	640	640	320	1280	ND
A/England/395/2008	22/09/2008	C2 \5	160	320	160	640	640	640	640	ND
A/Mauritius/L1/2008	21/11/2008	Cx \2	640	1280	320	1280	2560	1280	80	ND
A/Paris/458/2008	01/12/2008	C1 \1	640	1280	320	1280	5120	640	320	ND
A/Singapore/N593/2008	15/09/2008	Cx \1	1280	2560	640	1280	2560	1280	640	2560
A/Sweden/3/08	15/10/2008	MDCK2 \1	1280	1280	640	1280	1280	1280	320	640
A/England/687/2008	30/12/2008	MDCK P1 \2	640	1280	320	1280	2560	1280	320	640
A/Switzerland/1370948/2008	12/12/2008	MDCK2 \2	640	640	320	640	1280	320	160	320
A/Seoul/4436/2008	05/12/2008	MDCK-P2 \1	640	640	320	640	1280	640	320	640
A/Slovenia/60/2009	06/01/2009	MDCKx \1	640	640	320	640	1280	640	160	640
A/Slovenia/51/2009	05/01/2009	MDCKx \1	160	80	160	160	320	160	160	160
A/Baden-Württemburg/199/2008	15/12/2008	C4 \2	320	320	160	320	640	320	160	320
A/Parma/17/09	12/01/2009	I MDCK \1	80	80	160	160	160	80	160	160
A/Switzerland/1409281/2008	31/12/2008	cs \2	640	1280	320	640	1280	640	160	640
A/Leon/5/2009	13/01/2009	MDCK1 \1	1280	2560	640	1280	2560	2560	640	640
A/Clermont-F./1304/08	17/11/2008	P2MDCK \3	640	1280	320	1280	5120	1280	640	ND
A/Poitiers/1341/08	04/11/2008	Px+P1MDCK \3	640	640	320	1280	2560	640	320	ND
A/Paris/458/2008	01/12/2008	C1 \1	640	1280	320	1280	5120	640	320	ND
A/England/2/2009	24/12/2008	SIAT P1 \1	1280	1280	640	2560	5120	1280	640	ND
A/Norway/14/2009	01/01/2009	MDCK1 \2	640	1280	640	2560	2560	1280	640	ND
A/Thüringen/82/2008	19/12/2008	C1 \1	320	640	320	640	1280	320	160	ND
A/Baden-Württemburg/194/2008	19/12/2008	C3 \1	640	1280	320	1280	2560	640	320	ND
A/Torino/13/08	03/12/2008	MDCK1 \2	640	1280	320	1280	1280	640	320	640
A/Roma-ISS/3/09	12/01/2009	MDCK1 \1	640	640	320	1280	1280	640	320	640
A/Bratislava/144/2009	19/01/2009	MDCK1 \1	1280	2560	640	1280	2560	1280	320	640
A/Algeria/G155/2009	07/12/2008	Cx \1	640	1280	320	1280	2560	1280	640	640
A/Switzerland/1426835/2009	05/01/2009	cs \2	1280	1280	640	2560	2560	1280	640	1280
A/Switzerland/1431914/2009	07/01/2009	cs \2	2560	2560	1280	5120	5120	2560	640	1280
A/Tanger/536/09	13/01/2009	Cx \1	1280	1280	1280	5120	2560	2560	640	1280
A/Firenze/25/08	22/12/2008	MDCK1 \1	640	640	320	1280	1280	640	320	640
A/Roma-ISS/9/09	20/01/2009	MDCK1 \1	1280	2560	1280	2560	5120	2560	640	1280
A/Salamanca/10/2009	19/01/2009	MDCK1 \1	1280	2560	640	5120	5120	1280	640	1280
A/Bratislava/96/2008	19/12/2008	MDCK1 \1	1280	2560	1280	5120	5120	2560	640	2560
A/Meknes/518/09	13/01/2009	Cx \1	1280	2560	1280	5120	5120	2560	640	1280
A/Tanger/531/09	05/01/2009	Cx \1	1280	2560	1280	5120	5120	2560	640	1280
A/Soria/3/2009	09/01/2009	MDCK1 \2	1280	2560	640	2560	5120	1280	640	1280
A/Burgos/4/2009	09/01/2009	MDCK1 \2	1280	1280	640	2560	2560	1280	320	1280
A/Fes/521/09	31/12/2008	Cx \1	1280	2560	2560	5120	5120	5120	640	2560
A/Tanger/533/09	20/12/2008	Cx \1	1280	2560	1280	5120	5120	2560	640	2560
A/Torino/14/08	12/12/2008	MDCK1 \2	1280	2560	640	2560	5120	1280	320	1280
A/Roma-ISS/7/09	13/01/2009	MDCK1 \2	640	1280	320	1280	5120	1280	320	640
A/Roma/18/08	10/12/2008	MDCK1 \2	640	1280	320	1280	2560	1280	320	640
A/Athens/14/2009	13/01/2009	Cx \3	640	1280	320	1280	2560	1280	320	640
A/Athens/15/2009	13/01/2009	Cx \2	1280	2560	640	2560	5120	5120	640	1280
A/Algeria/G202/2009	23/12/2008	Cx \1	640	1280	640	1280	5120	1280	320	1280
A/Genoa/09/2009	06/01/2009	MDCK1 \1	640	1280	640	1280	2560	1280	320	1280
A/Genoa/10/2009	20/12/2009	MDCK1 \1	640	1280	640	1280	2560	1280	320	1280
A/Belgium/SP07/2009	14/12/2008	P1? \1	640	640	160	640	1280	640	320	640
A/Belgium/SP08/2009	21/12/2008	P1? \1	160	160	160	320	640	160	80	320
A/Algeria/G120/2009	23/11/2008	Cx \2	320	640	160	640	1280	640	160	320
A/Roma-ISS/2/09	12/01/2009	MDCK1 \2	320	640	160	640	1280	640	160	640
A/Parma/46/08	16/12/2008	MDCK1 \3	320	640	160	640	1280	640	160	640
A/Parma/52/08	31/12/2008	MDCK1 \3	160	80	80	160	160	80	80	160
A/Valladolid/12/2009	19/01/2009	MDCK1 \2	80	80	160	320	160	80	80	80
A/Salamanca9/2009	19/01/2009	MDCK1 \2	640	640	640	1280	1280	640	80	320
A/Barcelona/V258A/2009	11/01/2009	1 \1	1280	2560	640	2560	5120	2560	640	1280
A/Barcelona/V266A/2009	12/01/2009	1 \1	640	1280	640	1280	2560	1280	320	1280
A/Lisboa/43/2008	09/12/2008	MDCK 2 \3	1280	2560	640	2560	5120	2560	640	1280
A/Lisboa/01/2009	05/01/2009	MDCK 2 \2	1280	1280	640	2560	2560	2560	640	1280
A/León/13/2009	14/01/2009	MDCK 1 \2	1280	1280	640	2560	5120	2560	640	1280
A/Valladolid/14/2009	26/01/2009	MDCK 1 \2	1280	2560	640	2560	5120	2560	640	1280
A/Prague/63/2009	28/01/2009	MDCK2 \1	40	80	80	80	160	40	80	80

1. < = <40; ND = not done

Vaccine strains

Also tested by neutralisation

Table 6. Antigenic analysis of influenza A H3N2 viruses (Neutralisation)

			Neutralisation titre[1]					
			Post-infection ferret sera[2]					
Viruses	**Collection Date**	**Passage History**	**A/Well 1/04 F13/04**	**A/Bris 10/07 F18/07**	**A/Uru 716/07 F3/07**	**A/JHB 15/08 F22/08**	**A/Eng 394/08 F32/08**	**A/Bris 24/08 F3/09**
REFERENCE VIRUSES								
A/Wellington/1/2004	26/01/2004	E3 \1	**1280**	320	320	160	320	320
A/Brisbane/10/2007	06/02/2007	E2 \3	320	**2560**	5120	640	320	2560
A/Uruguay/716/2007	21/06/2007	spfck1, E3 \1	320	1280	**2560**	160	160	1280
A/Johannesburg/15/2008	26/06/2008	MDCKx \1	160	2560	2560	**5120**	320	ND
A/England/394/2008	22/09/2008	TC P2 \1	80	2560	2560	2560	**2560**	1280
A/Brisbane/24/08	23/06/2008	E5	160	1280	2560	ND	320	**2560**
TEST VIRUSES								
A/England/561/2008	04/12/2008	MDCK P1 \1	160	2560	2560	2560	640	ND
A/Finland/286/2008	08/12/2008	MDCK2 \1	160	2560	2560	2560	640	ND
A/England/85060527/2008	10/12/2008	SIAT P2 \1	160	1280	2560	1280	320	ND
A/Madagascar/3226/2008	23/09/2008	MDCK1 \1	80	1280	1280	1280	320	ND
A/Netherland/379/2008	24/09/2008	MK2 \1	160	1280	2560	2560	640	ND
A/Luxembourg/1325/2008	unknown	MDCKx \1	160	2560	2560	2560	320	ND
A/England/572/2008	08/12/2008	MDCK P1 \1	80	2560	2560	2560	640	ND
A/England/392/2008	11/09/2008	TC P1 \2	80	1280	1280	1280	320	ND
A/England/394/2008	22/09/2008	TC P2 \1	160	1280	2560	1280	320	ND
A/England/395/2008	22/09/2008	TC P2 \1	320	5120	5120	5120	160	ND
A/Thuringen/76/2008	26/11/2008	Cx \1	80	1280	1280	1280	320	ND
A/Norway/2118/2008	25/11/2008	MDCK \1	160	1280	2560	1280	320	ND
A/Mauritius/L1/2008	21/11/2008	Cx \1	320	5120	5120	5120	160	ND
A/Paris/130/2008	20/10/2008	C2 \1	160	640	1280	ND	320	1280
A/Egypt/59/2008	16/11/2008	C1 \1	160	1280	1280	ND	320	1280
A/Toulouse/1317/2008	14/11/2008	P3MDCK \1	160	1280	2560	ND	640	2560
A/Norway/2308/2008	15/12/2008	MDCK1 \1	320	2560	2560	ND	640	2560
A/Denmark/191/2008	13/11/2008	MDCK3 \1	160	1280	2560	ND	320	2560
A/Sweden/6/2008	08/12/2008	MDCK2 \1	160	1280	1280	ND	320	1280
A/Paris/458/2008	1/12/2008	C1 \1	320	2560	5120	ND	640	5120
A/HongKong/3197/2008	26/11/2008	MDCK2 \1	160	1280	1280	ND	320	1280
A/England/669/2008	30/12/2008	MDCK P1 \1	80	1280	1280	ND	320	1280
A/Norway/1/2009	30/12/2008	MDCK1 \1	160	640	1280	ND	320	1280

[1] Based on 50% plaque reduction compared to serum negative controls

Vaccine strains

[2] RDE treated

ND = not done

Agglutinated guinea pig RBCs only

Figure 4. Phylogenetic comparison of influenza H3N2 HA genes

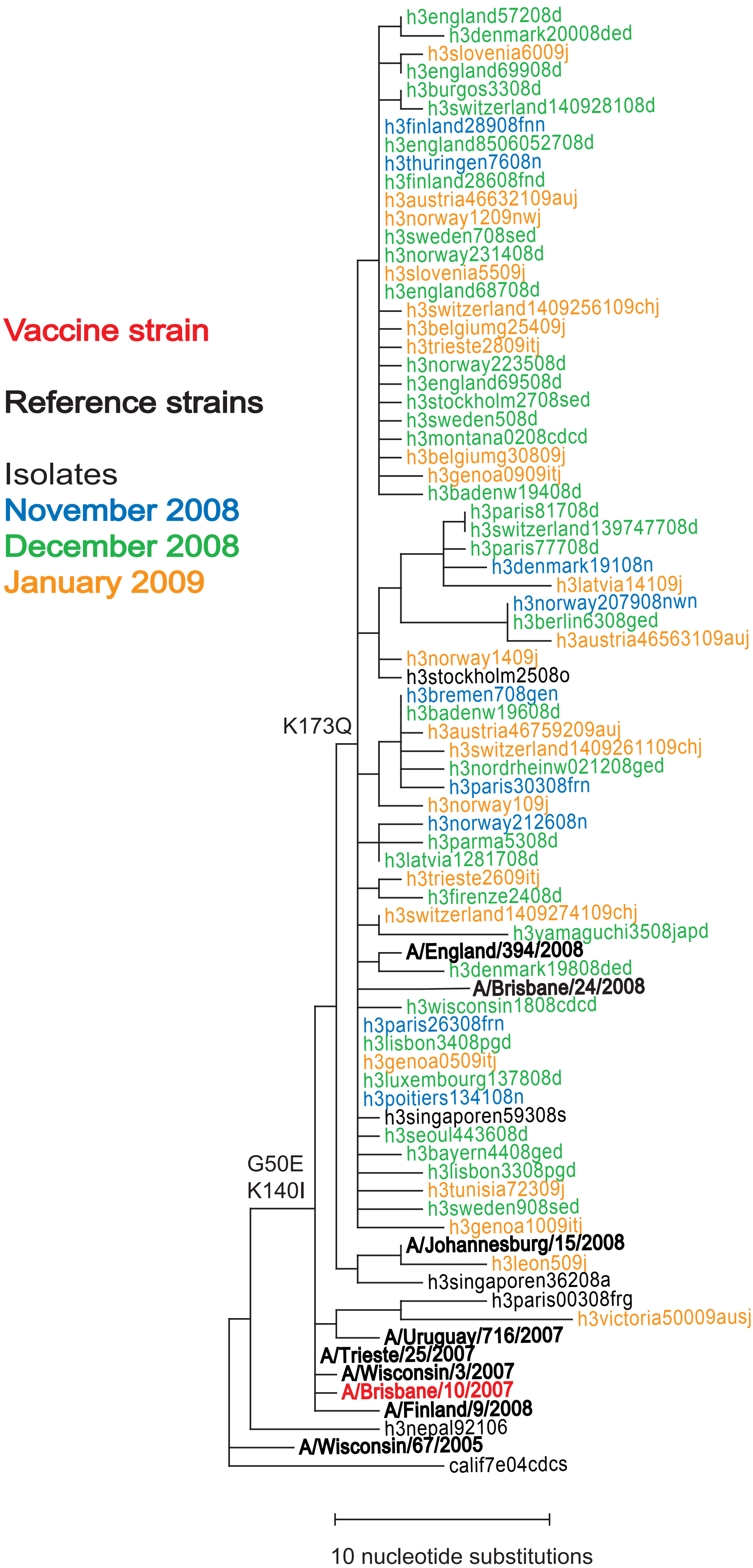


Figure 5. Phylogenetic comparison of influenza H3N2 neuraminidase genes

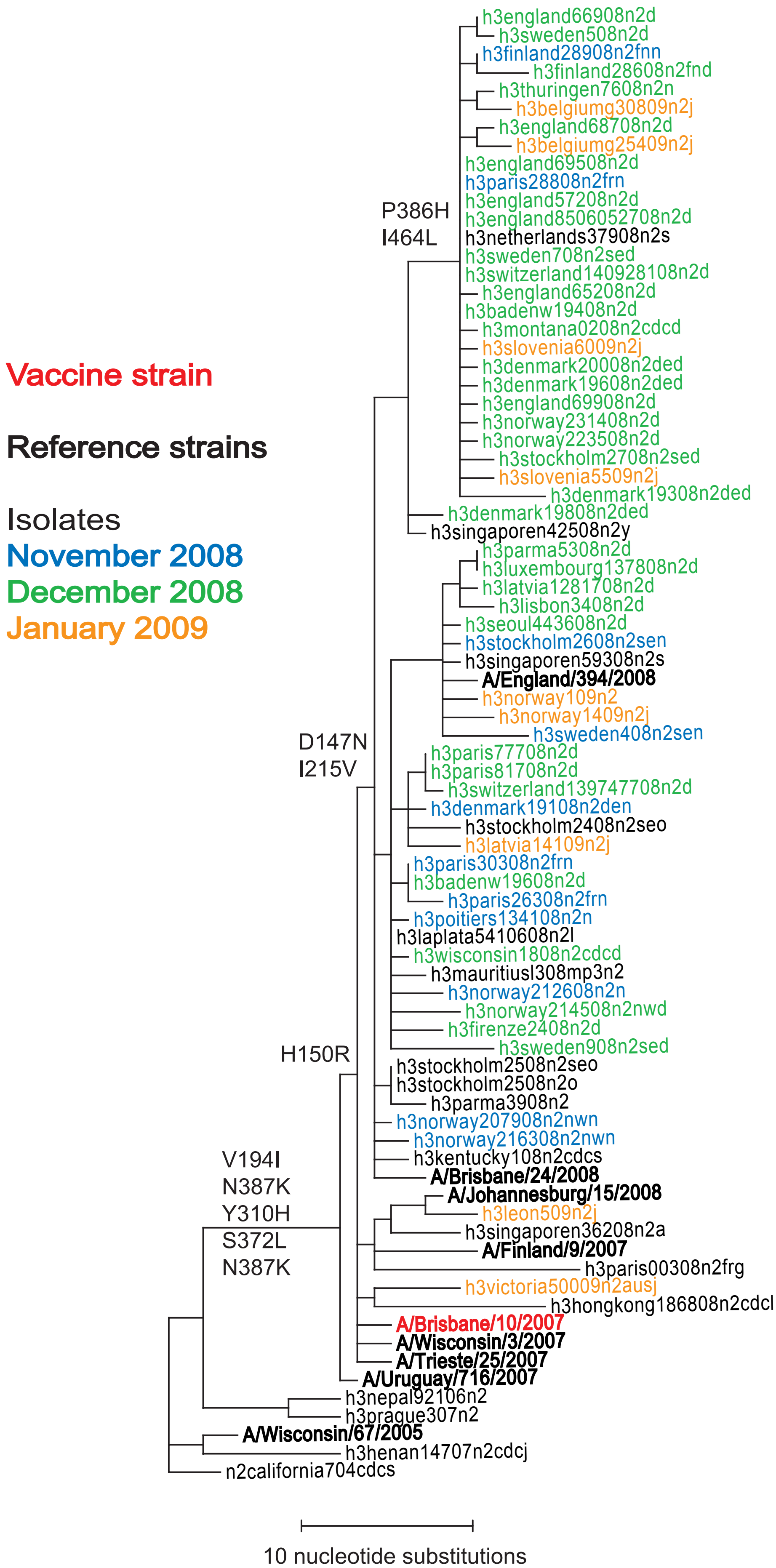


Table 7. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]								
				Post infection ferret sera							
			B/Shan[2] 7/97	B/Shan 7/97 F12/02	B/Bris 32/02 F2/03	B/HK 45/05 F15/05	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/Eng 393/08 F31/08	B/Bris 60/08 F2/09	B/Bris 33/08 AUS F1235
REFERENCE VIRUSES											
B/Shandong/7/97		Ex	1280	**160**	80	160	160	160	40	80	320
B/Brisbane/32/2002	02/07/2002	E2 \6	1280	160	**80**	80	160	160	20	40	320
B/HK/45/2005	07/02/2005	MDCK2 \4	1280	160	80	**160**	160	<	<	80	640
B/Malaysia/2506/2004	06/12/2004	E3 \3	1280	160	160	160	**320**	160	80	160	640
B/Victoria/304/2006	06/06/2006	E2 \2	1280	160	80	80	160	**160**	40	160	640
B/England/393/2008	29/08/2008	E1 \2	320	80	40	<	160	160	**320**	320	1280
B/Brisbane/33/2008	13/07/2008	E3 \1	640	80	40	<	160	160	320	320	**1280**
B/Brisbane/60/2008	04/08/2008	E4 \1	640	80	40	<	160	160	320	**320**	1280
TEST VIRUSES											
B/Hong Kong/1347/2008	01/09/2008	MDCK2 \1	1280	<	<	320	<	<	<	<	40
B/Hong Kong/1373/2008	08/10/2008	MDCK2 \3	2560	<	<	160	<	<	<	<	40
B/Agadir/235/09	11/12/2008	Cx \1	1280	<	<	160	<	<	<	<	40
B/Tanger/522/09	19/12/2008	Cx \1	5120	40	<	160	80	<	<	<	160
B/Tanger/438/09	30/12/2008	Cx \1	1280	<	<	160	<	<	<	<	80
B/England/457/08	31/10/2008	PLC / Egg P3 \1	160	40	40	<	80	80	320	160	1280
B/Texas/26/2008	24/11/2008	E3 \1	320	40	40	<	160	80	80	160	1280
B/Baden Wurttemberg/158/2008	06/10/2008	1.Pa(RKI) \1	640	<	<	<	<	<	<	80	640
B/Bayern/36/2008	23/10/2008	MDCK2 \1	640	<	<	<	<	<	<	ND	1280
B/Singapore/14/2008	03/11/2008	MDCK1 \1	1280	<	<	<	<	<	<	<	40
B/Belgium/G086/2009	19/11/2008	MDCK1 \1	1280	<	<	<	40	<	<	160	640
B/England/558/2008	01/12/2008	SIAT P1 \1	640	<	<	<	<	<	<	80	640
B/Bayern/40/2008	15/12/2008	MDCK2 \1	640	<	<	<	<	<	<	80	640
B/Bayern/39/2008	18/12/2008	MDCK2 \1	640	<	<	<	<	<	<	<	40
B/Niedersachsen/183/2008	23/12/2008	MDCK2 \1	640	<	<	<	<	<	<	80	320
B/England/711/2008	28/12/2008	MDCK P1 \1	640	<	<	<	<	<	<	80	640
B/Berlin/42/2008	29/12/2008	MDCK2 \1	640	<	<	<	<	<	<	80	640
B/Switzerland/1426895/2009	05/01/2009	cs \4	640	<	<	<	<	<	<	80	320
B/Barcelona/98/2008	06/11/2008	0 \1	320	<	<	<	<	<	<	160	640
B/Barcelona/99/2008	10/11/2008	0 \1	640	<	<	<	<	<	<	160	640
B/Barcelona/V362A/2009	14/01/2009	1 \1	640	<	<	<	<	<	<	160	640
B/Israel/2/09	11/01/2009	MDCKx \1	320	<	<	<	<	<	<	80	320
B/Israel/3/09	17/01/2009	MDCKx \1	320	<	<	<	<	<	<	40	320
B/Bratislava/108/2009	07/01/2009	1 \1	320	<	<	<	40	<	40	160	640
B/Volos/16/2009	09/01/2009	Cx \3	320	<	<	<	10	<	<	80	ND
B/Volos/68/2009	20/01/2009	Cx \3	160	<	<	<	10	<	<	80	ND
B/Athens/84/2009	23/01/2009	Cx \3	160	<	<	<	10	<	<	80	ND

1. < = <10; 2. hyperimmune sheep serum

ND = not done

Vaccine strain

Figure 6. Phylogenetic comparison of influenza B HA genes (Victoria lineage)

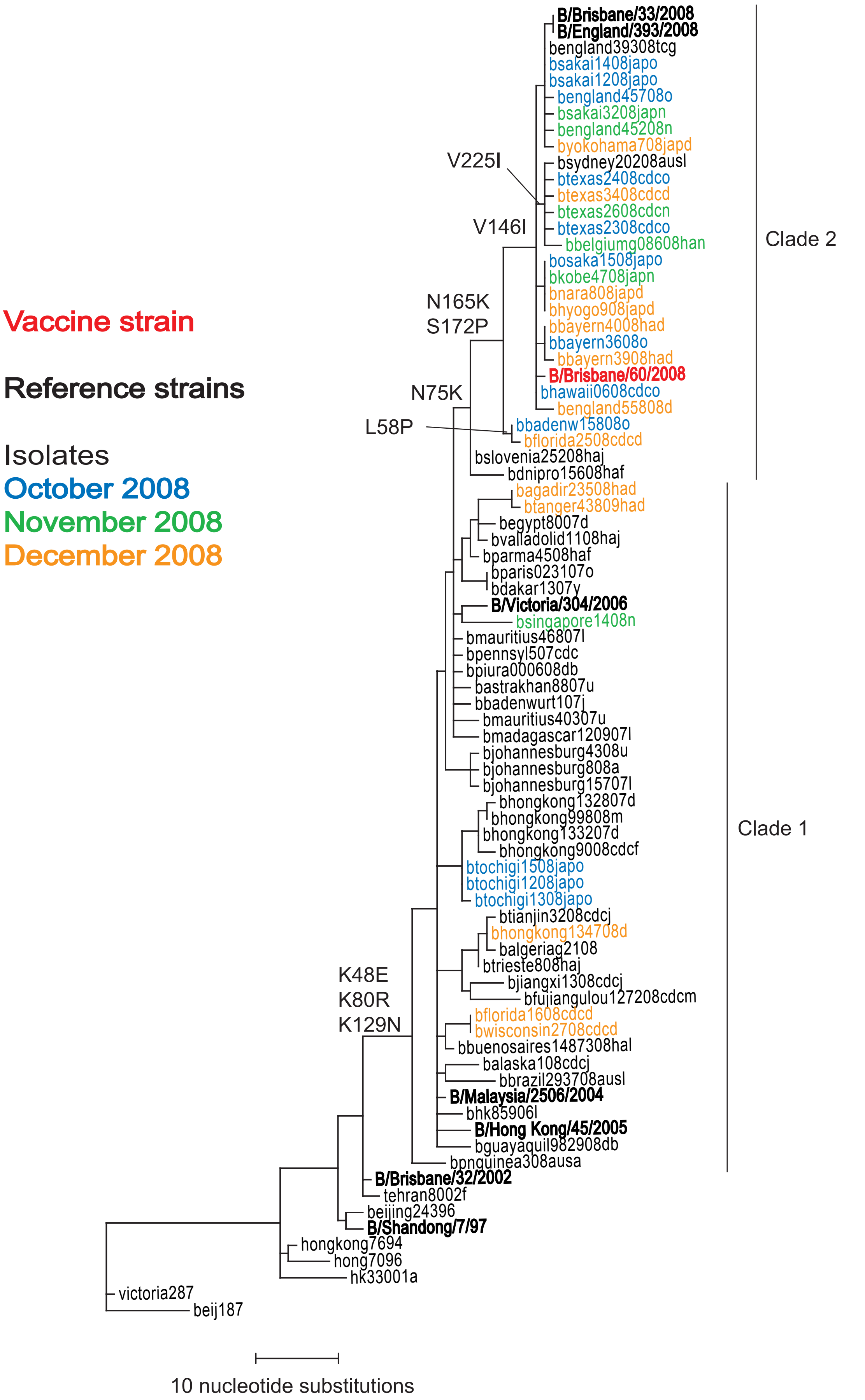


Figure 7. Phylogenetic comparison of influeza B neuraminidase genes

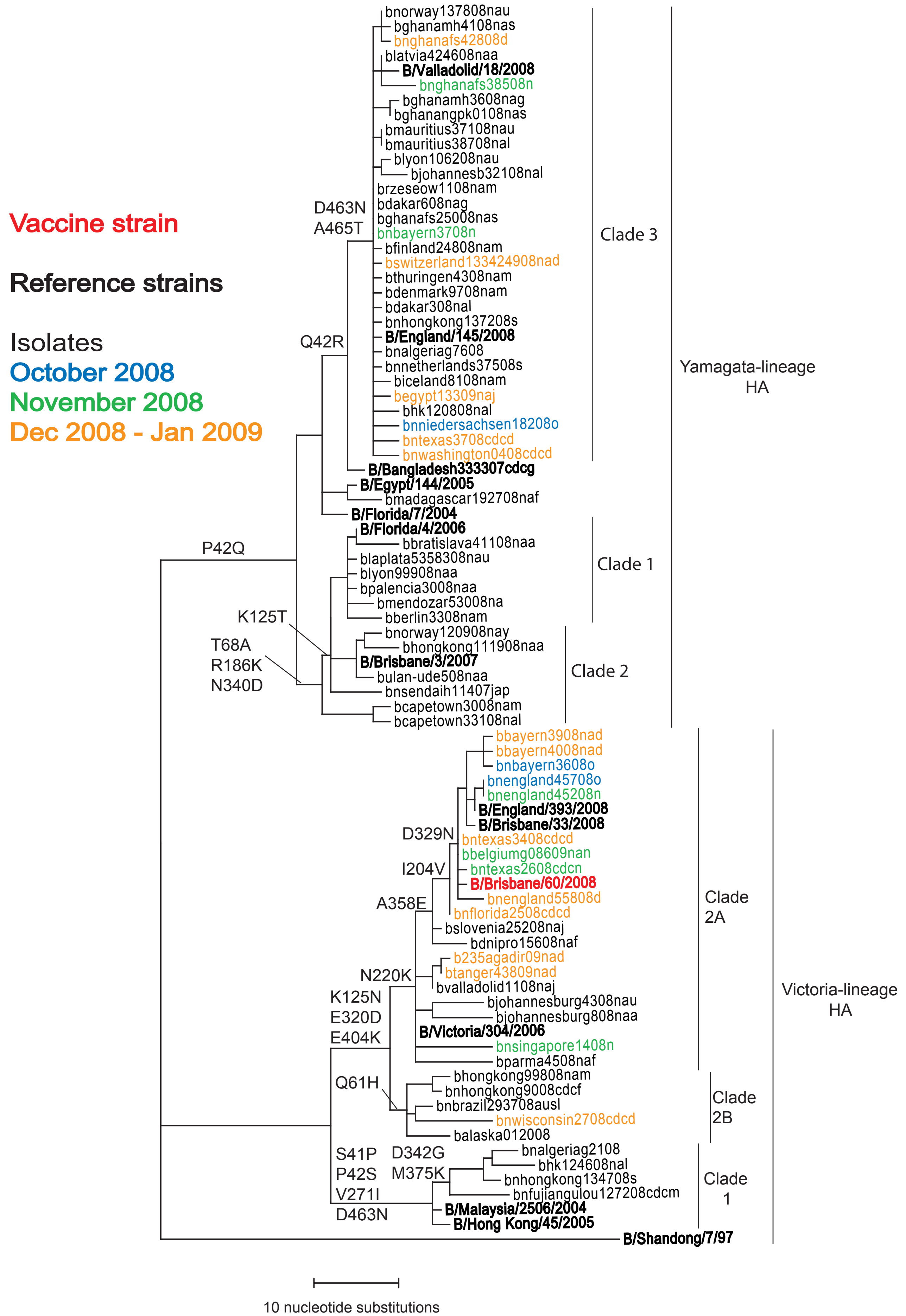


Table 8. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]							
				Post infection ferret sera						
			B/Fl[2] 4/06 Sh 478	B/Eg 144/05 F3/07	B/Fl 4/06 F19/07	B/Bris 3/07 F24/07	B/Barc 143/08 F11/08	B/Eng 145/08 F9/08	B/Vall 18/08 F10/08	B/Ban 3333/07 F21/08
REFERENCE VIRUSES										
B/Egypt/144/2005	01/05/2005	E2 \2	2560	**160**	320	640	10	40	20	160
B/Florida/4/2006	15/12/2006	E2 \1	5120	160	**640**	640	10	80	20	160
B/Brisbane/3/2007	03/09/2007	E2 \2	5120	160	640	**640**	10	80	20	160
B/Barcelona/143/2008	06/01/2008	MDCK1 \2	5120	80	160	320	**320**	160	320	320
B/England/145/2008		Ex \2	320	40	<	40	10	**160**	10	20
B/Valladolid/18/2008	18/02/2008	MDCK1 \2	2560	80	160	80	160	160	**320**	160
B/Bangladesh/3333/2007	19/08/2007	E4 \1	2560	80	160	160	20	40	20	**160**
TEST VIRUSES										
B/Dakar/06/2008	19/08/2008	MDCK2 /1	5120	160	640	640	320	640	320	640
B/Hong Kong/1355/2008	09/09/2008	MDCK2 \2	2560	80	160	160	160	160	160	320
B/Ghana/MH-41/2008	12/09/2008	MDCKx /1	2560	80	40	40	80	160	160	160
B/Hong Kong/1372/2008	25/09/2008	MDCK2 \1	5120	160	160	160	160	320	160	160
B/Ghana/FS301/2008	09/10/2008	MDCK2 \1	5120	80	160	320	320	160	320	320
B/Hong Kong/1374/2008	17/10/2008	MDCK2 \1	2560	80	160	320	160	160	160	160
B/Ghana/FS347/2008	31/10/2008	MDCK2 \1	2560	80	80	80	160	160	320	160
B/Ghana/FS351/2008	31/10/2008	MDCK2 \1	5120	80	160	80	320	160	320	320
B/Paris/236/2008	03/11/2008	C2 \1	640	40	<	<	10	80	40	40
B/Ghana/FS435/2008	12/12/2008	MDCK2 \3	2560	40	80	80	160	80	160	160
B/Switzerland/1334249/2008	04/12/2008	MDCK1 \1	2560	80	160	320	320	160	320	320
B/Egypt/133/2009	14/01/2009	Ex \1	160	<	<	<	<	20	<	10
B/Belgium/G212/08	24/12/2007	MDCKx \1	160	320	640	640	320	320	320	160
B/Switzerland/1334249/2008	04/12/2008	MDCK1 \1	2560	80	160	320	320	160	320	320
B/Bayern/37/2008	25/11/2008	2.Pa \1	640	80	160	40	40	40	40	160
B/Cameroon/08-177/2008	05/11/2008	MDCKx \1	2560	80	80	160	ND	80	160	160
B/Cameroon/08-229/2008	09/12/2008	MDCKx \1	2560	80	80	80	ND	80	160	320
B/Barcelona/51/2008	28/10/2008	0 \1	2560	160	160	<	ND	<	<	<
B/Norway/2054/2008	14/11/2008	MDCK1 \1	640	80	160	40	40	40	80	160
B/Pavia/3/08	03/11/2008	I MDCK \1	5120	160	320	320	320	320	640	320

1. < = <10; 2. hyperimmune sheep serum

Vaccine strains

ND = not done

Figure 8. Phylogenetic comparison of influenza B HA genes (Yamagata lineage)

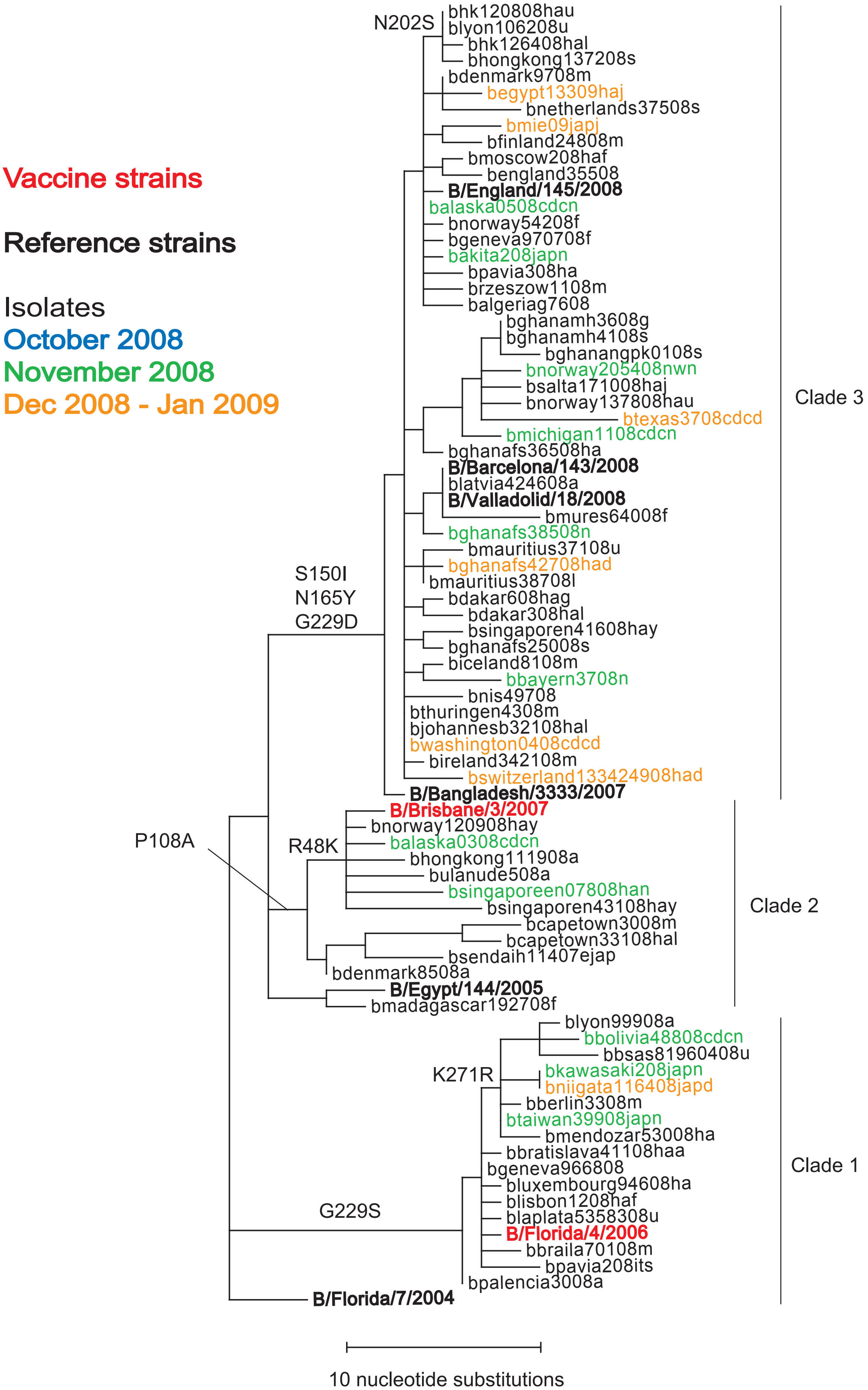


Table 9. Amino acid changes characteristic of recent influenza B sequence variants

HA Lineage	Representative strain	HA Clade	HA Changes	NA Clade	NA Changes
Yamagata[1]	**B/Florida/4/2006**	1	G229S K271R D478E	1	T68A K125T R186K N340D
	B/Brisbane/3/2007	2	R48K P108A	2	T68A K125T R186K I248V N340D
	B/England/145/2008	3	S150I N165Y G229D	3	Q42R D463N A465T
Victoria[2]	**B/Hong Kong/1347/2008**	1		1	D342G M375K
	B/Brazil/2937/2008	1		2B	P41S S42P Q61H K125N I271V E320D E404K N463D
	B/Brisbane/60/2008	2	N75K V146I N165K S172P (L58P) (V225I)	2A	P41S S42P K125N N220K I271V E320D A358E E404K N463D (I204V) (D329N)

1. Relative to B/Florida/7/04. 2. Relative to B/Malaysia/2506/04.